UCC 103 PRINCIPLES OF COMPUTING
CHAPTER ONE: OVERVIEW OF COMPUTERS

Objectives
At the end of the chapter the learner shall be able to;

i. Explain the evolution of computing technology and the technological advancement in computer architecture to current technologies
ii. Explain the characteristics of computers and how they are different from humans.
iii. Explain the different types of computers categorized based on size, price and capabilities
iv. Explain the fundamental difference between computer hardware and software

1.1 Basic Computer Concepts

The term computer comes from the word compute.

Definition

- A computer is an electronic device capable of executing instructions, developed based on algorithms stored in its memory, to process data fed to it and produce the required results faster than human beings.
- It is an electronic device which processes given data to derive the required and useful information. During the processing the computer has to perform various functions like, Accepting Instructions & data from the user, performing various arithmetic and Logical operations as per Instructions given and presenting the Information or Output to the user.

Other Definitions;

- The definition from the Merriam-Webster Dictionary: "one that computes; *specifically* : a programmable electronic device that can store, retrieve, and process data"
- A computer is a device that accepts data in one form and processes it to produce data in another form i.e information.
- An electronic device that takes data and instructions as an input from the users. Processes data and provides useful information known as output.

1.2. History of Computers

When the human race started doing some trade, it felt a need for a calculating device. The first calculating device, which was used 2000 years ago was called **abacus** and the improvements in the calculating device in that age were slow. The next change came after about 1600 years. Following this, the changes were frequent and the mechanical desk calculator was developed around 1800 A.D. In 1833, Prof. Charles Babbage, the father of the computer, developed a machine called **analytical engine** which was the vase for the modern digital computer.

The development of computers has followed different steps in the terminology used and these steps of technological differences are called as GENERATIONS in computer terminology. There are totally five generations of computers till today.

1.3 Generation of Computers

First Generation (1951-1958) by John W. Mauchly and J. Presper Eckert

The first generation of computers started with the UNIVAC I (Universal Automatic Computer) built by Mauchly and Eckert in 1951. It was sold to the U.S. Census Bureau. This machine was dedicated to business data processing example payroll and record keeping and not military or scientific purposes.

Characteristics

- Use of **vacuum tubes** in electronic circuits: These tubes controlled internal operations and were huge. As a consequence the machines were large.
- **Magnetic drum**; used as primary internal-storage medium: Electric currents passed through wires which magnetized the core to represent on and off states
- **Limited main-storage capacity:**
- **Slow input/output**, punched-card-oriented: Operators performed input and output operations through the use of punched cards.
- **Low level symbolic-language programming**: The computer used machine language which was cumbersome and accomplished through long strings of numbers made up of Zeroes and Ones. In 1952, Dr. Grace Hopper (University of Pennsylvania) developed a symbolic language called mnemonics (instructions written with symbolic codes). Rather than writing instructions with Zeroes and Ones, the mnemonics were translated into binary code. Dr. Hopper developed the first set of programs or instructions to tell computers how to translate the mnemonics.
- **Heat and maintenance problems**: Special air-conditioning and maintenance were required of the machines. The tubes gave off tremendous amounts of heat.
- **Applications**: payroll processing and record keeping though still oriented toward scientific applications than business data processing.

  Examples: IBM 650 UNIVAC I

Second Generation Computers (1959-1964)

Characteristics

- **Use of transistors** for internal operations: tiny solid state transistors replace vacuum tubes in computers. The heat problem was then minimized and computers could be made smaller and faster. The cost of the computers also reduced. Data access time was measured in micro-seconds.
- **Magnetic core** as primary internal-storage medium: Electric currents pass through wires which magnetize the core to represent on and off states. Data in the cores can be found and retrieved for processing in a few millionths of a second.
- **Increased main-storage capacity**: The internal or main storage was supplemented by use of magnetic tapes for external storage. These tapes substituted for punched cards or paper. Magnetic disks were also developed that stored information on circular tracks that looked like phonograph records. The disks provided direct or random access to records in a file.
- **Faster input/output**; tape orientation: Devices could be connected directly to the computer and considered "on-line". This allowed for faster printing and detection and correction of errors.
- **High-level programming languages** (COBOL,FORTRAN):These languages resembled English. FORTRAN (FORmula TRANslator) was the first high-level language that was accepted widely. This language was used mostly for scientific applications. COBOL (Common Business-Oriented Language) was developed in 1961 for business data processing. Its main features include: file-processing, editing, and input/output capabilities.
- **Increased speed and reliability**: Modular-hardware was developed through the design of electronic circuits. Complete modules called "breadboards" could be replaced if malfunctions occurred, or the machine "crashed". This decreased lost time and also new modules could be added for added features such as file-processing, editing and input/output features.
- **Batch-oriented applications**: billing, payroll processing, updating and inventory files: Batch processing allowed for collection of data over a period time and then one processed in one computer run. The results were then stored on magnetic tapes.

Examples:IBM 1401*(most popular business-oriented computer. Honeywell 200 CDC 1604

Third generation computers (1965-1970)

Characteristics

- **Use of integrated circuits:** The use of integrated circuits (*Ics*) replaced the transistors of the second-generation machines. The circuits are etched and printed and hundreds of electronic components could be put on silicon circuit chips less than one-eighth of an inch square.
- **Magnetic core and solid-state main storage**: Greater storage capacity was developed. They have higher main memory capacity, reliable and increased processing power (have the capability of holding more than one set of instructions and operate on them) than the second generation computers.
- More flexibility with input/output; **disk-oriented:**
- **Smaller size and better performance and reliability**: Advances in solid-state technology allowed for the design and building of smaller and faster computers. The small circuitry that resulted improved the processing speed i.e. 10 times the past. The speed of these machines was described in nanoseconds (1/1,000,000,000 of a second).
- Extensive use of **high-level programming languages**: The software industry evolved during this time. Many users found that it was more cost effective to buy pre-programmed packages than to write the programs themselves. The programs from the second generation had to be rewritten since many of the programs were based on second generation architecture.
- **Emergence of minicomputers:** The mini computers offered many of the same features as the mainframe computers only on a smaller scale. These machines filled the needs of the small business owner.
- **Remote processing** and **time-sharing** through communication: Computers were then able to perform several operations at the same time. Remote terminals were developed to communicate with a central computer over a specific geographic location. Time sharing environments were established.
- Availability of operating-systems (software) to control I/O and do tasks handled by human operators: Software was developed to take care of routine tasks required of the computer freed up the human operator.
- **Applications** such as airline reservation systems, market forecasting, credit card billing: The applications also included inventory, control, and scheduling labor and materials. Multitasking was also accomplished. Both scientific and business applications could be run on the same machine.

Examples: IBM System/360 NCR 395 Burroughs B6500

Fourth Generation (1970-)

Characteristics:

- Use of Large Scale Integrated (LSI) and very Large Scale Integration (VLSI) circuits which housed hundreds of transistors and very large IC which are between 200,000 to 400, 000 in one IC.
- Smaller and cheaper
- Increased storage capacity and speed. Memory used includes magnetic disc and optical disc. Memory size expanded up to several MB and speed was 10 times faster.
- Modular design and compatibility between equipment
- Special application programs
- Versatility of input/ output devices

- Increased use of minicomputers
- Introduction of microcomputers. This generation marked the origin of microcomputers in use today
- Applications: mathematical modeling and simulation, electronic funds transfer, computer-aided instruction and home computers. Internet Explosion.
- Enabled creation of Local Area Networks (LAN) and Wide Area Networks (WAN) which facilitates connection and communication of hundred of computers located across multiple locations.
- Operating systems includes; DOS and Windows. Use of GUI – Graphical User Interface – a user friendly interface that provides ICONS and Menus to the user to interact with the various computer applications.
- No air conditions

Fifth generation computers (1990-current)

The Fifth Generation Computer Systems project (FGCS) was an initiative by Japan's Ministry of International Trade and Industry, begun in 1982, to create a "fifth generation computer" which was supposed to perform much calculation using massive parallel processing.

- The design of these computers was based on VLSI (very large scale integration) technology, the micro chip technology that gave rise to the smaller computers known as the micro computers in use today.
- These computers are used in networking .examples of micro computers are IBM PCs BBC micro etc. the micro computers are usually described as PCs or stand alone or desktop computers because they were designed primarily to serve single person at a time.
- The fifth generation is still a state of the art technology that relies on predictions and further technological refinements.
- It has been predicted that such a computer will be able to communicate in natural spoken language with its user;
- store vast knowledge databases;
- search rapidly through these databases,
- making intelligent inferences and drawing logical conclusions; and
- process images and ‘see’ objects in the way that humans do.
- Reliable and vault tolerance

Summary

Research shows that the trend in computer technology revolution is that there is;

- Continual decrease in computer size
- Improved speed and power processing
- Decrease in computers and the related facilities cost
- Number of components in computer per circuit (IC) greatly increased over 500,000 physical elements e.g. transistors, capacities, diodes etc per chip(IC).

1.4. Generations of Computer Languages

1) First Generation – machine language
   - Machine language is the only programming language that the computer can understand directly without translation. It is a language made up of entirely 1s and 0s.
2) Second Generation Assembly mnemonic Language

- Assembly languages use mnemonic operation codes and symbolic addresses in place of 1s and 0s to represent the operation codes. A mnemonic is an alphabetical abbreviation used as memory aid. This means a programmer can use abbreviation instead of having to remember lengthy binary instruction codes.
- Assembly language uses symbolic addressing capabilities that simplify the programming process because the programmer does not need to know or remember the exact storage locations of instructions or data

3) Third Generation Language
   - Third generation languages, also known as high-level languages. Most high level languages are considered to be procedure-oriented, or Procedural languages, because the program instructions comprise lists of steps, procedures, that tell the computer not only what to do but how to do it. High-level language statements generate, when translated, a comparatively greater number of assembly language instructions and even more machine language instructions.
   - A language translator is required to convert a high-level language program into machine language. Two types of language translators are used with high level languages: compilers and interpreters.
4) Fourth Generation Language
   - High level Language also. They are non-procedural languages, so named because they allow programmers and users to specify what the computer is supposed to do without having to specify how the computer is supposed to do it.
   - Five basic types of language tools fall into the fourth generation language category.
     a) Query languages
     b) Report generators.
     c) Applications generators.
     d) Decision support systems and financial planning languages.
     e) Some microcomputer application software.
5) Fifth Generation Language
   - These languages did not appear until the 1990s, and have primarily been concerned with Artificial Intelligence and Fuzzy Logic. The programs that have been developed in these languages have explored Natural Language (making the computer seem to communicate like a human being).
   - Natural languages widely known as fifth generation languages, are designed to make a computer to behave like an expert and solve problems. The programmer just needs to specify the problem and the constraints for problem solving. Natural languages such as LISP and PROLOG are mainly used to develop artificial intelligence and expert systems.

1.5. Characteristics of Computers

Computers are so versatile that they have become indispensable to engineers, scientists, business executives, managers, administrators, accountants, teachers and students. Modern computers posses certain characteristics and abilities peculiar to them.

1) Perform complex and repetitive calculations rapidly and accurately.
2) Store large amounts of data and information for subsequent manipulation.
3) Compare items and make decisions.
4) Provide information to the users in many different forms
5) Automatically correct or modify the parameters of a system under control
6) Converse with the users interactively
7) Receive and display audio and video signals
8) Speed – a computer is a very fast machine. It can perform in a very few seconds the amount of work that a human being can do in a year if he/she worked day and night doing nothing else.

9) Accuracy – the computer accuracy is consistently high.
10) Diligence – computers are free from monotony, tiredness and lack of concentration etc. It can therefore work for hours without creating an error. For example if 10 million calculations are to be done, a computer will do the tenth million calculations with exactly the same speed and accuracy as the first one.
11) Versatility – a computer performs various tasks with ease. The computer perform three basic operations
    a) It is capable to access and accept information through various input-output devices from the user.
    b) It performs basic Arithmetic and Logic operations on data as desired.
    c) It is capable to generate the desired output in the desired form.

    Example i.e. it can search for a letter, the next moment prepare an electricity bill, and write a report next then do an arithmetic calculation all with ease.
12) Power of remembering – a computer can store and recall any information due to its secondary storage capability.
13) No intelligence Quotient (IQ) – a computer cannot make its own decisions and has to be instructed on what to do.
14) No feelings – computers are devoid of emotions. They have no feelings or instincts and none possesses the equivalent of a human heart and soul.

1.6. Limitation of Computer

2. Computer does not work on itself, it requires set of instructions to be provided, else computer (Hardware) is waste.
3. Computer are not intelligent, they have to be instructed about each and every step which they have to perform
4. Computers cannot take decisions on its own, one has to program the computer to take an action if some conditional prevail.
5. Computers, unlike humans cannot learn by experience.

1.7. Advantages and Disadvantages of Computers

Advantages - Application Areas

Computers can be used in the following application areas

1) Data processing (Commercial use).
2) Numerical computing (Scientific use).
3) Text (word) processing (Office and Education)
4) Message communication (E-mails)
5) Image processing (Animation and industrial use).
6) Voice recognition (Multimedia)

Disadvantages

1) Unemployment
2) Computer Crime
3) Data Security
4) Violation of Privacy
5) Waste of Time and Energy
6) Health Risks
7) Impact on Environment

1.8. Classification of Computers

Computers can be classified many different ways –by type, size, purpose, function, and/or by processing capacity.

1.8.1. Basic Type:

- Digital computers: process data that is represented in the form of discrete values (eg. 0,1,2,3,..) by operating on its in steps. Discrete values occur at each step in the operation
- Analog Computers: akin to measuring instruments such as thermometers and voltmeters with pointers on circular dials. They process data in the form of electrical voltages, which are variable like the positions of a pointer on a dial.
- Hybrid: computers that have the combined features of digital and analog computers.

1.8.2. Classification by purpose

This give rise to;

- Special purpose computers: designed for a particular job only; to solve problems of a restricted nature, example computers used in digital watches, programmable pocket calculators, petrol pumps etc.
- General purpose computers: designed to solve a wide variety of problems.

1.8.3. Classification by function

Classification by the work the computer is used for. This result in;-

- Word processor: used for production of documents.
- Home Computers: designed for domestic use, have limited capacity etc. eg, computer games
- Desktop computers: designed for use on an office desktop
- Workstation: have more features and capabilities than the desktop computers, such as inbuilt capabilities for interconnection and operation in conjunction with other computers.
- Lap-top: smaller than a PC and portable.
- Embedded computers: inserted within some device or system and are not accessed directly.

1.8.4. Classification by size

This gives rise to;

Mainframe computers

A mainframe is another giant computer after the super computer and can also process millions of instruction per second and capable of accessing billions of data .They are physically very large in size with very high capacity of main memory. This computer is commonly used in big hospitals, air line reservations companies, and many other huge companies prefer mainframe because of its capability of retrieving data on a huge basis. They can be linked to smaller computers and handle hundreds of users they are also used in space exploitation. A large number of peripherals can be attached to them. They are expensive to install.

Minicomputers

They are smaller than the main frame but bigger than minicomputers. They support concurrent users. They can be used as servers in companies. They are slower and less costly compared to mainframe computers but more powerful, reliable and expensive than micro computers.

Micro computers

They are of advanced technology i.e. the micro chip era based on large scale integration that confines several physical components per small elements thumb size IC, hence the size reduced. It is the smallest of the three computers. They are usually called personal computers since they are designed to

be used by individuals. The micro chip technology has enabled reduction of size of computers. Microcomputers can be a desktop, laptop, notebooks, or even palmtop

- **Notebook computer** An extremely lightweight personal computer. Notebook computers typically weigh less than 6 pounds and are small enough to fit easily in a briefcase. Aside from size and portability. Notebook computers use a variety of techniques, known as *flat-panel technologies,* to produce a lightweight and non-bulky display screen.
- **Desktop Computer** is an independent personal computer that is made especially for use on a desk in an office or home. The term is used mainly to distinguish this type of personal computer from portable computers and laptops, but also to distinguish other types of computers like the server or mainframe.
- **Laptop** A small portable computer light enough to carry comfortably, with a flat screen and keyboard that fold together. Laptops are battery-operated, often have a thin, backlit or sidelit LCD display screen, and some models can even mate with a docking station to perform as a full-sized desktop system back at the office.
- **Palmtop** A small computer that literally fits in your palm. Compared to full-size computers, palmtops are severely limited, but they are practical for certain functions such as phone books and calendars. Palmtops that use a pen rather than a keyboard for input are often called *hand-held computers* or *PDAs.* Because of their small size, most palmtop computers do not include disk drives.
-

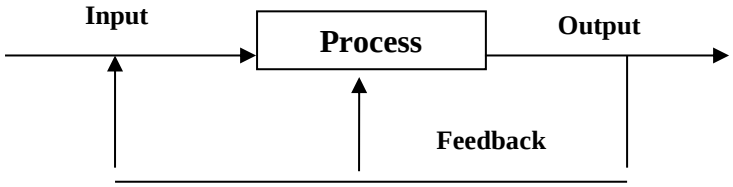
1.8.5. Classification by processing power

Super Computers

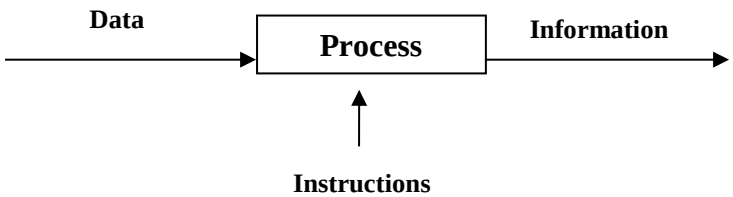
They are very large in size and use multiple processors and superior technology. Super computers are biggest in size, the most expensive in price than any other is classified and known as super computer. It can process trillions of instructions in seconds. This computer is not used as a PC in a home neither by a student in a college. Governments specially use this type of computer for their different calculations and heavy jobs. Different industries also use this huge computer for designing their products.

1.9. Main Activities of a Computer

A ***computer*** is an electronic device that operates under the control of a set of instructions that is stored in its memory unit. A computer accepts data from an input device and processes it into useful information which it displays on its output device.



The relationship between Computer, Data and Information can be depicted as follows;



1.10 Components of a Computer

A ***computer*** is an electronic device that operates under the control of a set of instructions that is stored in its memory unit. A computer accepts data from an input device and processes it into useful information which it displays on its output device. Actually, a computer is a collection of hardware and software components that help you accomplish many different tasks. *Hardware* consists of the computer itself, and any equipment connected to it. *Software* is the set of instructions that the computer follows in performing a task.

A computer has to main components;

1. Hardware

Computer hardware refers to the physical components of a computer. The primary / basic hardware components of a computers are;

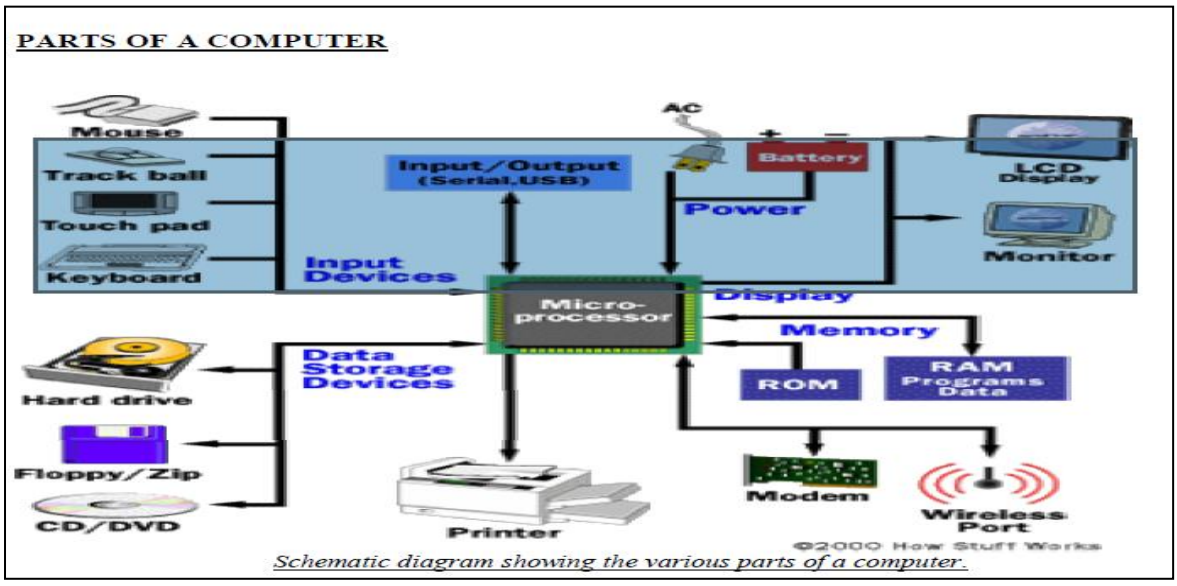
1) Input: devices used to capture and send data and instructions to the processing component / unit of the computer
2) Storage: used to store the data and instructions until needed to be worked on. Also used to hold the processed information until required for output. There are two types of storage; main memory(storage) and secondary storage
3) Control: used to fetch instructions from the main storage, interprets them, and issues the necessary signals to the components making the computer system. It directs all hardware operations necessary in obeying instructions.
4) Processing: processing is done by the Arithmetic Logical Unit (ALU) which obeys the instructions and performs the required arithmetic operations, and logical operations
5) Output: devices used to output the processed data (information). This includes, screen, printer etc.

2. Software

A set of programs associated with the operations of a computer

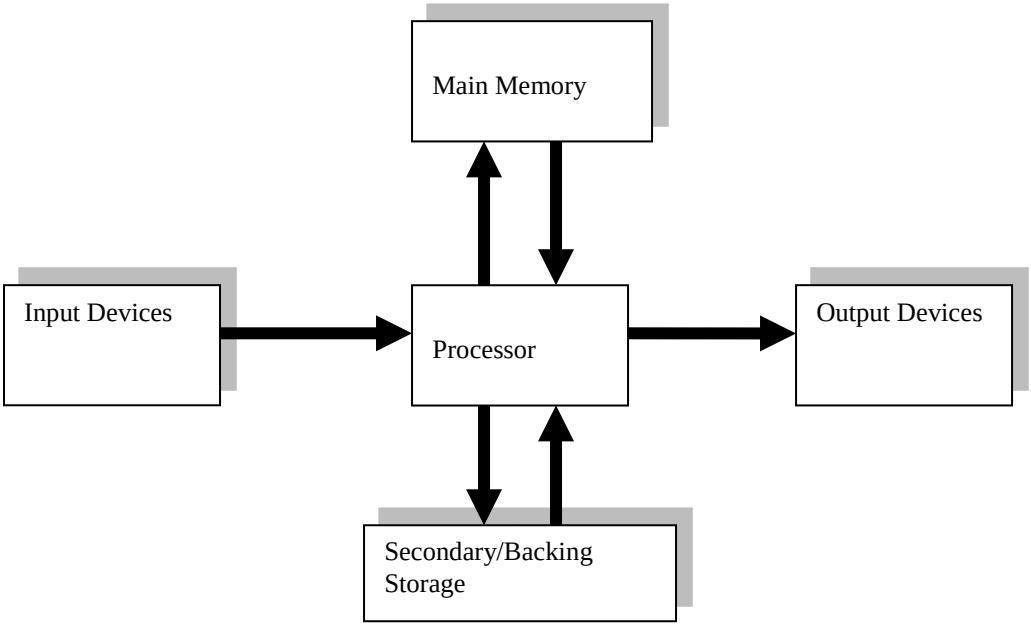
a) System Software
b) Application Software

The two components (hardware, software) will be discussed later in other chapters.



Schematic diagram showing the various parts of a computer.

The Relationship can be depicted as follows;



Summary: Definition of terms

- Data: basic facts that have not been processed and have no meaning
- Information: processed data, useful and have meaning.
- Program: set or series of instructions that is written in the language of the computer, which specifies processing / tasks that the computer is to carry out on data.
- Hardware is the name given to all the physical devices found in a computer system.
- Software is the general term used to describe all the various programs that may be used on a computer system together with their associated documentation.

Computer Performance

Features that affect the performance of the computer include:

a) microprocessor
b) Operating System
c) RAM
d) disk drives
e) display
f) input/output ports

1.7 Chapter Review Questions

1. The second generation of computers used
   (a) Vacuum tubes (b) Capacitors (c) Transistors (d) Integrated circuits
2. The third generation of computers used
   (a) Vacuum tubes (b) Capacitors (c) Transistors (d) Integrated circuits
3. The **analytical engine** was the vase for the modern digital computer in which year was is developed?
   (a) 1833 (b) 1933 (c) 1923 (d) 1893
4. Which one of the following types of computers is commonly used in offices
   (a) Supercomputers (b) Mainframe (c) Mini computer (d) Micro computer
5. Computers have continued to decrease in size but the processing power has increased. **True or false**?